

## RADIOACTIVITY

In 1896, Henry Becquerel discovered a previously unknown type of radiation, which is emitted by atomic nuclei. As we know today, this radioactive radiation can consist of  $\alpha$  particles (helium nuclei),  $\beta^+$  or  $\beta^-$  particles (positrons or electrons),  $\gamma$  quanta (high-intensity radiation), or other particles. It can be found in natural radioactive elements such as uranium and radium, which are among the heaviest elements. Today, many lighter radioactive nuclei are produced in “artificial” nuclear reactions.

### RADIOACTIVE DECAY

Nucleons (protons and neutrons) stick together because of a strong attractive short-range force (the *strong interaction*), which prevents the nucleus from falling apart under the repulsive Coulomb force between the protons. Stable nuclei are possible for certain combinations of proton and neutron numbers only. These two numbers are more or less equal with a slight excess of neutrons for heavy nuclei.

Unstable isotopes disintegrate into a stable nucleus (usually not of the same element), possibly passing by several unstable intermediate nuclei. The decay is accompanied by the emission of radiation.

The number of disintegrations per second is proportional to the number of active nuclei:  $\Delta N = -\lambda \cdot N \cdot \Delta t$

The *decay constant*  $\lambda$  is characteristic for the decay of a given nucleus. For short time intervals,  $\lambda$  is the probability that the nucleus will decay during the time  $\Delta t$ .

The relative change in the number of active nuclei is constant for equal intervals of time. This is a characteristic property of an *exponential decay*. It follows that the number of active nuclei after the time  $t$  is given by

$$N(t) = N_0 \cdot e^{-\lambda t}$$

where  $N_0 = N(0)$  is the number of active nuclei at time  $t = 0$ .

The *half life* of a decay is related to the decay constant:

$$T_{1/2} = \frac{\ln 2}{\lambda}$$

The half lives of different radioactive elements span a range from smallest fractions of a second to more than  $10^9$  years.

### ACTIVITY

The *activity* of a radioactive sample is the number of disintegrations per second:

$$A(t) = \left| \frac{\Delta N}{\Delta t} \right| = \lambda N$$

$[A] = 1 \text{ disintegration per second} = 1 \text{ Bq (Becquerel)}$

Activity is also measured in a unit called *curie* (Ci), in honour of Marie (1867 – 1934) and Pierre (1859 – 1906) Curie:  $1 \text{ Ci} = 3.7 \cdot 10^{10} \text{ Bq}$ .

## PROBABILITY DISTRIBUTIONS

It is impossible to predict the time when a given nucleus will disintegrate. Instead, we only know the probability that it will do so in the next second. As a result, measurements of the activity of a radioactive sample are always subject to statistical fluctuations, which have nothing to do with measurement errors.

The probability that from a sample of  $N$  active nuclei exactly  $x$  will disintegrate during the time is given by the *binomial distribution*:

$$P_N(x) = \binom{N}{x} p^x q^{N-x}$$

where  $p = \lambda \Delta t$  is the probability that a nucleus will decay during the time  $\Delta t$  and  $q = 1 - p$  is the probability of the complementary event.

For weak samples ( $N p^2 \ll 1$ ), the binomial distribution can be approximated by the *Poisson distribution*

$$P_N(x) = \frac{p^x N^x e^{-pN}}{x!}$$

under certain conditions also by the *normal distribution*

$$P_N(x) = \frac{1}{\sqrt{2\pi N p q}} e^{-\frac{(x - pN)^2}{2N p q}}$$

(see "Formulae, Tables, Concepts", p. 121 ff.). In the latter case the *standard deviation* corresponds to  $\sqrt{N}$ .